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Review

Application of Wearable Sensors in Parkinson's Disease: State of the Art

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Abstract: (1) Background: Wearable sensors have emerged as a promising technology in the management of Parkinson's disease (PD). These sensors can provide continuous and real-time monitoring of various motor and non-motor symptoms of PD, allowing for early detection and intervention. In this paper, I review current research on the application of wearable sensors in PD, focusing on gait, tremor, bradykinesia, and dyskinesia monitoring.(2) Methods: this involved a literature search that spanned the 2000–2024 period and included the following keywords: "wearable sensors", "Parkinson's Disease", "Inertial sensors", "accelerometers", ''gyroscopes", ''magnetometers", "Smartphones", and "Smart homes". (3) Results: Despite favorable outcomes from the early development of inertial sensors, like gyroscopes and accelerometers in smartphones, the application of wearable sensors is still restricted because there are no standards, harmonization, or consensus for both clinical and analytical validation. As a result, several clinical trials were created to compare the effectiveness of wearable sensors with conventional evaluation methods in order to track the course of the disease and enhance the quality of life and results. (4) Conclusions: wearable sensors hold great promise in the management of PD and are likely to play a significant role in future healthcare systems.

Keywords: wearable sensors; Parkinson's disease; inertial sensors; smartphones; smart homes

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1. Introduction

Parkinson's disease (PD) is a complex neurodegenerative disorder that presents with a wide range of motor (tremors, bradykinesia, postural instability) and non-motor symptoms, including autonomic disorders and behavioral and cognitive deficits [1]. In the late stage, levodopa-related motor complications include(a) off effects, where the duration of the drug's beneficial effect decreases with each dose; (b) delayed on/off fluctuations, where the time between taking levodopa and the clinical effect is either delayed or nonexistent; and (c) random on/off fluctuations [2]. Traditional methods of assessing PD symptoms rely on subjective clinical evaluations (e.g., diaries, questionnaires), which can be prone to bias and variability [3,4].

Flexible electronics have the power to transform a number of different sectors in the age of artificial intelligence, including wearable skins, healthcare, the human–machine interface (HMI), and the Internet of Things (IoT) [5–7]. Flexible wearable medical electronics, in particular, are capable of measuring clinical parameters, like blood pressure, body temperature, respiration, and electrical activity, and integrate seamlessly with human tissues over an extended period of time. This allows for the real-time provision of basic health data for big data-driven medical treatments [8,9].

Indeed, the emergence of wearable sensor technology has revolutionized the monitoring and management of PD by enabling the objective, continuous, and remote assessment of key motor symptoms. Wearable sensors are small, lightweight devices that can be worn on the body to capture various physiological parameters, such as movement, tremors, gait, balance, and cognitive and sleep patterns [10,11]. These sensors utilize a range of supporting networking technologies, including accelerometers, gyroscopes, magnetometers, and electromyography, to collect data on motor and non-motor symptoms in PD patients [12]. While some methods have suggested using many sensors on various body parts, more contemporary methods often utilize sensors on the foot, wrist, fingers, cellphones with wrist adapters, or devices that resemble watches [12–16]. Additionally, these methods have been combined with threshold approaches, automatic classification algorithms, and deep learning techniques [7–11] to offer inexpensive and non-invasive remote surveillance systems. The data collected by wearable sensors can be analyzed in realtime or stored for later review, providing valuable insights into disease progression, treatment efficacy, and quality of life through developments in machine learning [12,13] (Figure 1).

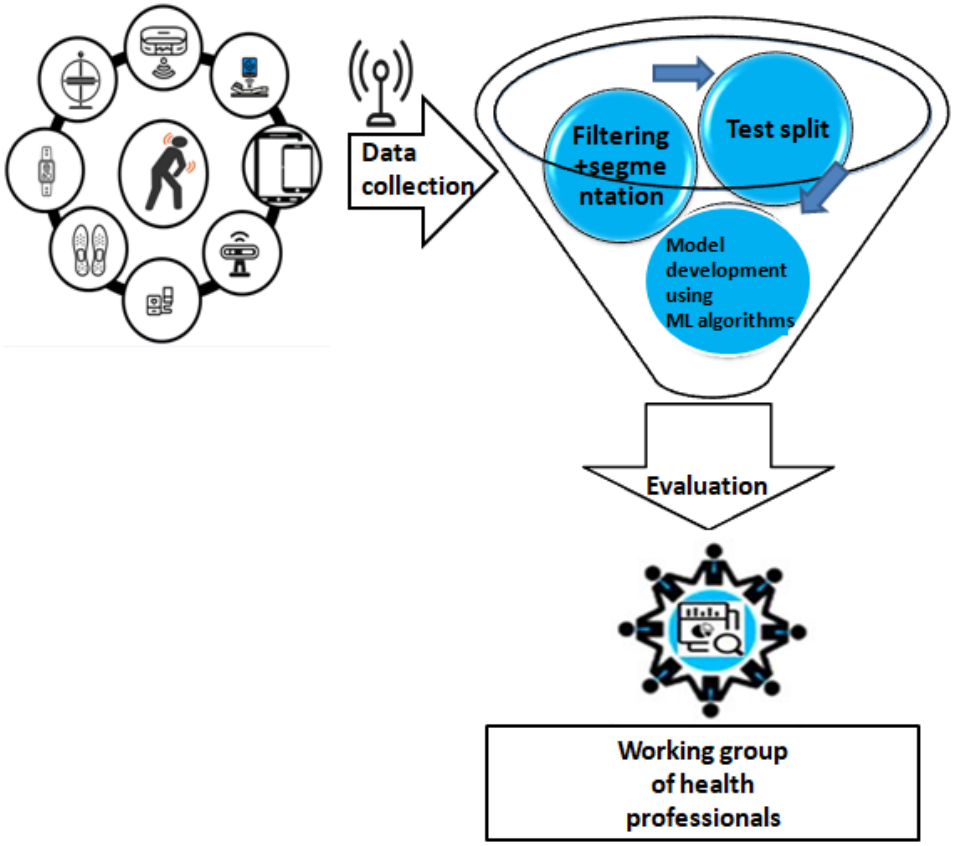


Figure 1. An overview of the sensors for PD and their data collection and analysis by health professionals (modified from Bougea [16]). To detect body-motion-related data, such joint angle, rotation during movement, and acceleration and speed of movement, wearable sensors are affixed to various body areas(e.g., wrist-worn accelerometers, smartwatches, forearm and shoe accelerometers, smart insoles), as well as tools for measuring motor and non-motor symptoms in various body parts (e.g., finger-tapping keyboards, facial monitoring cameras, smartphones, smart mattress).The development process for wearable machine learning-based software requires specific operations related to machine learning applications, such as data collection, filtering and segmentation, and test splitting. A working group of health professionals (including biostatisticians, neurologists, physiotherapists, psychologists, nurses) should evaluate the reliability of machine learning algorithms in transforming sensor data into meaningful physiological metrics.

Previous reviews focused on wearable technologies [14], such as smartphone apps [15] and inertial sensors, using machine learning models to predict motor/non-motor symptoms of PD [12,16,17]. This narrative review aimed to provide a more comprehensive overview of the current applications by adding novel technologies and providing a deeper discussion of the benefits, challenges, and future directions of wearable sensors in the management of PD.

2. Materials and Methods

2.1. Data Collection

Articles published during the 2000–2024 period were reviewed, with a focus onwearable sensors in PD patients. PubMed, Scopus, and Embase were searched from 2000 to 2024.In more detail, the following Boolean search string was used: ((*Parkinson*) OR (*Motor symtoms*) OR (*PDMonitor*®) OR (*PKG*®*system*) OR (*STAT-ONTM*), OR (*KinesiaONETM*), OR (*FeetMe*® *Monitor*), OR (*DynaPort7TM*), OR (*Mobility Lab*™), OR (*APDM*®), OR (*Universal Asynchronous Receiver–Transmitter*) OR (*accelerometers*), OR (*gyroscopes*), OR (*magnetometers*), OR "Smartphones", OR "Smart homes", OR (*soft multifunctional neurological E-skin*), OR (*Triboelectricnanogenerators*), OR (*Non-motor symptoms*)) AND (*Wearable*) AND (*Digital*) AND (*Wearable*).

No additional filters were applied in the literature search.

2.2. Inclusion and Exclusion Criteria

The inclusion criteria were original research papers, reviews, and meta-analyses that have been published in peer-reviewed publications in the English language. The exclusion criteria were as follows:

1. Books, book chapters, peer-reviewed papers, and publications issued as "letters", "comments", "perspectives", "case reports", or "surveys".
2. Studies of non-English literature.
3. Research on illnesses other than PD.
4. Research that collected data without the use of portable sensors or wearable technology.
5. Research that demonstrated the outcomes of a contest, challenge, or initiative.
6. The majority of the studies concentrated on activities unrelated to PD motor symptoms.
7. Research without human participants.

The snowballing procedure was carried out to screen the references of each selected article for potential extra papers to cover the current key evidence, followed by a full-text evaluation of pertinent research. For the narrative synthesis, my initial categorization was based on the benefits of wearable sensors in the management of PD, followed by a discussion of limitations and potential future directions.

3. Results

3.1. Wearable, Inertial Sensors and FDA-Approved Devices

Small electrical devices called inertial sensors, sometimes referred to as motion sensors, monitor acceleration, angular velocity, and sometimes the intensity of magnetic fields. An object's orientation and motion in three dimensions are determined using these measurements. Accelerometers, gyroscopes, and magnetometers are examples of inertial sensors that are frequently used to record a variety of motion-related data. Wearable motion sensors have shown promise in the everyday life monitoring of PD patients due to their compact size, low weight, and low energy requirements [16–20]. Table 1 summarizes research on wearable PD sensors.

Table 1. Selected studies of wearable sensors for PD.

Authors (Publication)	Year	Sensor Type	Sample Characteristics	Setting	ML Algorithms/ Motor Parameter Measurements	Grade of Agreement Between Kinematic Data * and Clinical Evaluations	Main Limitations
Hoffman et al. [21]	2011	Triaxial gyroscopes and adaptive filters	11 PD pts	Laboratory	UPDRS/ angular velocity signal	The angular velocity signal was less consistent than those obtained from controls.	Small sample and the lack of follow-up.
Heldman et al. [22]	2012	Heel-wor n motion sensor	22 non-DBS pts 20 DBS pts	Home	MDS-UPDRS	Strong correlation coefficient of 0.86.	Small sample and the lack of follow-up.
Mera et al. [23]	2013	KinetiSense wrist-worn motion sensor (3 orthogonal accelerometers and 3 orthogonal gyroscopes)	42 pts with PD	Home	UPDRS mAIMS PDYS-26	Correlation coefficients between at least one kinematic feature and corresponding UPDRS scores were greater than 0.75 for six of the seven items. Kinematic features improved ($p < 0.05$) from DBS-off to DBS-on for three UPDRS items.	Wrong positioning of motion sensors and thevisual rating scales' logarithmic structure and their decreased sensitivity to barely perceptible variations at elevated severities.
Weiss et al. [24]	2014	Body-fixed sensor (3D accelerometer) on lower back	107 PD pts	Home	Walking quantity (e.g., steps per 3days) and quality (e.g., frequency-derived measures of gait variability)	No significant correlations between the UPDRS motor score and the acceleration-derived gait measures.	This approach cannot be applied to identify fall risk among patients with PD.
Ossig et al. [25]	2016	PKG	24 PD pts	Home	UPDRSIII motor score Patients' diaries BDI	Moderate-to-high correlation between patients' diaries and PKG's quantitative motor status.	Limited correlation betweenthe hour-to-hour diary and the PKG recordings.
Pérez-López et al. [26]	2016	Single belt-worn accelerometer	1st database:92 PD pts (validation cohort) 2nd database:10 PD pts	Home	Indoor walking test Outdoor walking test FoG provocation test Dyskinesia test False positive test	95% specificity and 39% sensitivity for moderate PD. 95% specificity and 93% sensitivity for severe and mild trunk dyskinesias.	Physical limitations of the PD patients and the usability characteristics of the system.
Samà et al. [27]	2017	Waist-worn triaxial accelerometer	12 PD pts of mild PD (HY $\leq$ 2.5)	Home	UPDRS	>90% accuracy for detection of bradykinesia.	Small sample and method dependence on gait analysis.
Bayés et al. [28]	2018	REMPARK System: sensor +smartphone	41 pts with moderate-to-severe idiopathic PD	Home	UPDRSIII motor score Patients' diaries	88% specificity (i.e., accuracy in recognizing on states) and 97% sensitivity whenidentifyingoff states.	Small sample, short observation period, and restricted evaluation of motor status (walking or dyskinesia).
Lee et al. [29]	2018	Shoe-type IMU sensor+a motion capture system comprising nine infrared cameras for 1min of treadmill walking	17 PD pts	Clinic	MDS-UPDRS MMSE/ linear accelerations, cadence, left step length, right step length, left step time, and right step time	Outstanding compatibility between the shoe-type IMU and the motion capture systems and motor parameters.	Small sample andtreadmill walking not easy to perform.

Table 1. *Cont.*

Authors (Publication)	Year	Sensor Type	Sample Characteristics	Setting	ML Algorithms/ Motor Parameter Measurements	Grade of Agreement Between Kinematic Data * and Clinical Evaluations	Main Limitations
Pulliam et al. [30]	2018	Wrist and ankle motion sensors recorded by video	13 PD pts	Clinic	UPDRS-III	Clinical evaluations of video recordings were in agreement with algorithm scores for tremor, bradykinesia, and dyskinesia (ROC area > 0.8).	Small sample and the need for home enviroment and more longer monitoring periods.
Wan et al. [31]	2018	Smartphone accelerometer	42 pts of early-stage PD	Laboratory	UPDRS ML classificational algorithms	Among all MLs, DMLP captured the accelerometer data from the smartphone the best.	Small sample size and a lack of real PD patients available to create the model and produce longitudinal, continuous smartphone data.
Buongiorno et al. [32]	2019	Microsoft Kinect v2 sensor	16 PD pts	Laboratory	MDS-UPDRS ANN SVM	AAN:89.4% accuracy for 9 PD features and 95.0% for 6 features when rating PD severity. SVM: 87.1% accuracy for 2 PD features.	Small sample size, lack of follow-up, and the need for home setting to validate the results.
Isaacson et al. [33]	2019	Kinesia360: wrist and ankle sensors	39 PD pts	Home	UPDRS Kinesia360-derived motor scores by time for the EG PDQ-39 PAM-13	Clinical decision-making was improved when objective, ongoing motor symptom monitoring was used in addition to conventional therapy.	Small sample size. The study was not powered to detect small changes in the Kinesia variables. The data may not reflect overall function at home (only "off" periods).
Mazzetta et al. [19]	2019	Wearable system	7 pts with moderate PD	Laboratory	Real-time FoG detection using gyro and sEMG MDS-UPDRS 7m TUG test FoG questionnaire	Prompt identification of the FoG episode and, for the first time, the automated differentiation of the phenotypes associated with fall risk.	Small sample and need for living-like environment for validation.
Xiong et al. [34]	2020	Vocal vector autoencoders	188 PD pts with and without vocal disorders	Laboratory	6 supervised ML algorithms	95% accuracy, 97% sensitivity, and 98% specificity of the WSO–sparse autoencoder–LDA in distinguished the PD and healthy samples.	Lack of longitudinal data, need for expert healthcare professionals for AI.

Table 1. *Cont.*

Authors (Publication)	Year	Sensor Type	Sample Characteristics	Setting	ML Algorithms/ Motor Parameter Measurements	Grade of Agreement Between Kinematic Data * and Clinical Evaluations	Main Limitations
Hadley et al. [35]	2021	Smartphone-basedKinesiaU™	16 pts with mild PD		Electronic diary symptom ratings for tremor, slowness, and dyskinesia	Significant recorded changes ($p < 0.05$) in symptom ratings over time after implementing a new treatment on hourly, daily, and weekly timelines.	Not clinically meaningful regarding the week-to-week changes andthe short follow-up periods.
van Kersbergen et al. [36]	2021	Microsoft Kinect V2 sensor	19 PD pts		Camera-derived data MDS-UPDRS	Significant differences between patients and controls for step length and average walking speed.	Small sample and the lack of follow-up.
Parati et al. [37]	2022	FeetMe®insoles GaitRite®electronic pressure-sensitive walkway	25 mild-to-moderatePD pts	Controlled clinical environ-ment	UPDRSpart III Spatiotemporal gait parameters	The two systems had significant convergence and mild-to-excellent associations.	Lack of severely affected PD group and age-matched healthy controls, gait waveforms could lead to falsely undetected stride, and the need for a home setting.
Safarpour et al. [38]	2022	3 inertial sensors	31 mild-to-moderate PD pts	Laboratory	MDS-UPDRS PIGD	Prediction of MDS-UPDRS rigidity and PIGD scores for walking bouts, gait speed, and postural sway.	Examination only forthe on state, small sample, and a lack of home examination of balance.
Adams et al. [39]	2023	Smartphone and smartwatch	82 pts with early, untreated PD	Home	MDS-UPDRS Trail-making test part A MoCA Symbol digital visuospatial working memory	Moderate-to-very substantial correlations ($0.36 < r < 0.79$) between the gait measures obtained from the smartphone and wristwatch and similar metrics from the research-grade wearable sensors.	Missing information, wearing the watch in different directions, unfamiliarity with certain tasks, the uniform research population, and concerns over the significance of the measurements.
Antoniniet al. [40]	2023	PDMonitor®	65 PD pts	Hospital	UPDRS part III AIMS CRS (all measures inoff state)	96% accuracy/97% specificity/85% sensitivity foroff; 99% accuracy/99% specificity/82% sensitivity for dyskinesia time;96%accuracy/ 98%specificity/83% sensitivity for FoG.	Need for home settingand using remote monitoring in clinical practice.

Table 1. *Cont.*

Authors (Publication)	Year	Sensor Type	Sample Characteristics	Setting	ML Algorithms/ Motor Parameter Measurements	Grade of Agreement Between Kinematic Data * and Clinical Evaluations	Main Limitations
Keogh et al. [41]	2023	McRoberts Dynaport MM + waist-worn device	20 PD pts	Home	Interviews CRS Rabinovich questionnaire	The device was seen as accessible and comfy. the grade of agreementnot reported.	Small sample, lack of interview expertise, and did not assess computer literacy or health of the pts; therefore, we are unable to take it into consideration when interpreting these findings.
Bailo et al. [42]	2024	G-Walk sensor by a semi-elastic belt	22pts with mild PD	Laboratory	MDS-UPDRS II +III MoCA 6MWT TUG Leg asymmetry index Rigidity index Tremor index BMI PIGD	Moderate-to-strong correlation of reliable gait measures with at least 1 clinical scale.	Lack of group with severe PD and of cut-offs due to the small sample of healthy controls. Difficult to define the clinical meaning of the moderate/high discriminative ability of the system.

AIMS: Abnormal Involuntary Movement Scale, ANN: Artificial Neural Network, BDI: Beck's Depression Inventory, BMI: Body Mass Index, CRS: comfort rating scale, DB: digital biomarker, DBS: deep brain stimulation, FoG: freezing of gait, mAIMS: modified Abnormal Involuntary Movement Scale, ML: machine learning, MMSE: Mini-Mental State Examination, MoCA: Montreal Cognitive Assessment, MDS-UPDRS: Movement Disorder Society–Unified Parkinson's Disease Rating Scale, 6MWT: 6-Minute Walk Test, PAM-13: 13-Item Patient Activation Measure, PD: Parkinson's disease, PDQ-39: 39-Item Parkinson's Disease Questionnaire, PDYS-26: 26-Item Parkinson's Disease Dyskinesia Scale, PIGD: Postural Instability Gait Difficulty, PKG: Parkinson's kinetigraph, pts: patients, SVM: Support Vector Machine, TUG: Timed Up and Go; * tremor, bradykinesia, and dyskinesia.

By detecting unexpected fluctuations in acceleration, accelerometer sensors enable the monitoring of posture and movement. Gyroscopes, on the other hand, detect rotational movements and angular velocity. Magnetometers evaluate heading and orientation tracking by measuring the orientation and recording alterations in the magnetic field. The wrist accounts for 18.2% of all accelerometer, gyroscope, and magnetometer attachments, with the lower leg (13.6%), waist (12.1%), upper leg (9.1%), and foot (6.1%) following in order of frequency [43]. When used together, these tools provide unbiased information on changes in posture and motor function. The most widely used sensors are a magnetometer, gyroscope, and tri-axial accelerometer. An inertial measurement unit (IMU), or three-dimensional construct with linear and angular velocity, is produced when they are integrated into a single device [44–46]. Small devices enable extended free walking outside labs for assessing PD motor symptoms, particularly due to the Hawthorne observation effect, which suggests that free-living activities with varying complexities and distractions may reduce attention.

Inertial sensors can differentiate PD from healthy controls based on movement patterns, enabling machine learning algorithms to identify specific features. They can also monitor PD progression, capture changes in movement patterns, and provide insights into the disease course [44–52]. Longitudinal data can inform treatment decisions and assess efficacy, such as tremor severity and gait disturbances.

Inertial sensors were used in observational studies to advance digital progression biomarkers for PD, such as the Personalized Parkinson Project [53], the Cincinnati Cohort Biomarker Program (CCBP) [54], the Wearable Assessment in The Clinic and at Home in PD (WATCH-PD) [39,55], the Oxford Parkinson Disease Centre (OPDC) discovery cohort [56], and the ALAMEDA study [57]. These devices objectively measure the efficacy of new interventions by capturing changes in motor symptoms duringtrials. By documenting variations in motor symptoms during a trial, these tools are used to objectively assess the effectiveness of novel therapies, such as medicine, physical therapy, and deep brain stimulation. Inertial sensors can better assess the efficacy of novel treatments and provide insightful information about reactions to therapy by delivering continual and quantitative data. Inertial sensors are frequently employed in clinical trials to evaluate the main objectives pertaining to the degree of motor symptom severity and functional improvement. For instance, gyroscopes record improvements in gait metrics, whereas accelerometers detect changes in the intensity of tremors [39,53–55]. Experts can ascertain whether the therapy significantly affects motor symptoms and functional capacities by examining this data. Inertial sensors may offer secondary outcome measurements in addition to primary endpoints. These gadgets can provide a thorough understanding of a person's motor function and general mobility by recording information on movement patterns, activity levels, and postural changes. The effect of the treatment on routine tasks and quality of life—two crucial secondary goals in PD research—can be evaluated using these data. With everything taken together, inertial sensors are essential to the planning and implementation of PD clinical trials because they offer precise and quantitative assessments of functional skills and motor symptoms [39,52–56]. The accuracy and dependability of trial results could be increased by this technique, which would eventually result in the creation of more potent therapies.

NICE recommends wearable devices, like PDMonitor, PKG, STAT-ON, Kinesia360, and KinesiaU, for the remote monitoring of PD to inform treatment, as they belong to class I or IIa medical devices [58] (Table 2).

Table 2. Overview of commercial wearable sensors for PD monitoring.

Wearable Sensors' Commercial Names	Study Numbers, PD Populations (Sizes, Ages, HYs, Disease Durations) *	Clinical/ LaboratoryTools	Main Outcomes	Medical Class/FDA Authorization (Yes or No)	Cost Per Visit
PD Monitor®	Two studies (sample size: 61–65 patients) withage range: NA	Patients' diaries AIMS UPDRS FoG index Postural instability score Gait impairment score	Strong associations with video-recorded UPDRS and AIMS for bradykinesia and dyskinesia; very high for OFF-time (accuracy: 99% resting tremor, 96% FoG).	CE mark class IIa /no	EUR 210
PKG®	26 studies, large sample size (max sample size: 3288 patients), and age range: 30–80 yrs; disease duration range: de novo (2 months)—23 years; HY: 1–4 (but disease data have not been reported for all studies)	Patients' diaries AIMS UPDRS	Moderate correlation with bradykinesia, high correlation for dyskinesia, and tremor.97.1% sensitivity differentiating fluctuators vs. non-fluctuators.	CE mark class Iia /no	USD225
STAT-ON®	9 studies (12–75) withage range: 59–83; disease duration: 5–18 yrs; HY: 1–4	Patients' diaries UPDRS	3 studies with good accuracy for ON/OFF fluctuations detection when compared with patients diaries; 2 studies (12–75 patients) with moderate/high correlation vs. UPDRS-III and gait, respectively; one study with moderate correlation with trunk/leg dyskinesia; one study (39 patients) with fair agreement for dyskinesia and FoG but low agreement for motor fluctuations when compared with UPDRS IV.	CE mark class Iia /no	EUR 215
Kinesia360®	10 studies (13–60 patients) withage range: 46–85 yrs; disease duration: 2–31 yrs; HY: 1–4	Patients' diaries UPDRS TETRAS	High correlation for resting/postural tremor, dyskinesia, and bradykinesia vs. physician-based video recordings.	CE mark, class I/yes	GBP 224

Table 2. *Cont.*

Wearable Sensors' Commercial Names	Study Numbers, PD Populations (Sizes, Ages, HYs, Disease Durations) *	Clinical/ LaboratoryTools	Main Outcomes	Medical Class/FDA Authorization (Yes or No)	Cost Per Visit
DinaPort	3 studies (40–176 patients) withage range: 41–81; disease duration: 1–14 yrs; HY: 1–4	MDS-UPDRS	MDS-UPDRS correlated with several gait parameters and misstep with history of falls.	CE mark class II in US and classI in EU/yes	USD203,23
FeetMe Monitor Insoles®	One study (25 patients) withage (mean): 69 yrs; disease duration: 4 yrs; HY:2	GaitRite® gait parameters	High correlation for gait parameters between pressure-sensing insoles and the electronic walkway.	CE-marked class Im/no	USD175,54
APDM®	30 studies (max sample size: 198 PD patients) withage range: 52–78 yrs; disease duration: 1–19 yrs; HY: 1–4	Gold-standard laboratory metrics (GaitRite Mat, Optical MoCap, Force Plate) andclinical scales (UPDRS-III, PIGD score)	Moderate correlation with physicians'base scales (UPDRS-III, PIGD score). High correlation with FoG episodes longer than 2–5 and >5 s. Able to differentiate fallers vs. non fallers.	Class II/yes	USD 2399

AIMS: Abnormal Involuntary Movement Scale, HY: Hoehn and Yah. Medical class: Im—device with low risk (m for measure); IIa—with low to medium risk, installed on the body for 30 min up to 60 days. FDA: Food and Drug Administration, FoG: freezing of gait, MDS-UPDRS: Movement Disorder Society–Unified Parkinson's Disease Rating Scale, PIGD: postural instability and gait disorders. * if information on clinical features is not reported in the table, it is not reported in the literature. QUEST: Quebec User Evaluation of Satisfaction with assistive Technology questionnaire, TETRAS: Essential Tremor Rating Scale, yrs: years.

The PDMonitor® is a class IIa medical device designed to monitor patients with PD. It consists of five IMU sensing devices placed on the wrists, shins, and waist: (1) a SmartBoxis used to collect, process, and upload data to the cloud; (2) five sensing/monitoring devices each with internal data storage (up to 7 days) that can transfer data when docked into the SmartBox; and (3) accessories for attaching the devices to the patient. The system was used to assess 65 PD patients in the hospital and at home for at least 7 h/day [40]. The device calculated the probability of the patient being off based on their gait, tremors, FoG, and bradykinesia.

The PKG®system, first FDA-cleared in 2014, is a medical device for PD patients. It captures motor symptoms, monitors medication administration, detects device removal, and monitors movement during sleep. The watch, attached to the wrist, collects data for6–10 days and has a rechargeable battery and accelerometer. A mathematical algorithm produces the raw motion data, which are translated into bradykinesia and dyskinesia scores, which generate a graphical report. The device also measures motor complications, including fluctuations and dyskinesia scores. A summary score that combines variations in bradykinesia and dyskinesia scores to give the Fluctuations and Dyskinesia Score (FDS) is indicative of motor problems. The Percent Time in Tremor (PTT), a summary score for the ambulatory assessment of tremor, and the Percent Time Immobile (PTI), which shows the percentage of time the patient was completely immobile and aids in characterizing prolonged immobility, daytime sleep, and somnolence, are additional and distinct scores. The device showed high sensitivity and specificity in identifying tremors and distinguishing between fluctuators and non-fluctuators vs. the gold-standards, i.e., the AIMS score (dyskinesia) and UPDRS-III (bradykinesia) [59,60].

The STAT-ON™ is a wearable sensor that uses real-time machine learning algorithms to measure inertial signals. It weighs 86g and consists of two ultralow triaxial nano-accelerometers and two microcontrollers using Bluetooth Low Energy wireless technology. A single sensor places the system above the waist given that it is quite near to the mass center of the body, and several motions, at least partially, are reproduced there. This sensor records gait, step length, FoG, falls, and trunk or neck dyskinesia. The device was successful at identifying trunk dyskinesia, FoG, and falls with a high sensitivity>93%and specificity75–98% [27]. The sensor is user-friendly and allows for the input of events, like medication use.

The KinesiaONETM is a class I medical device that evaluates motor signs of PD using an iPad mini loaded with KinesiaONE software, an electronic diary, a wearable sensor, and a charge pad [59]. Three orthogonal accelerometers and three orthogonal gyroscopes, which measure linear acceleration and angular velocity, respectively, are part of the motion sensor. To measure motor symptoms, such as tremor, bradykinesia, dyskinesia, gait, and freezing of gait, the sensor is worn on the index finger while doing particular activities. Based on the Essential Tremor Rating Assessment Scale (TETRAS) for essential tremor and the UPDRS for Parkinson's disease, the mobile application offers guidance and instructions for completing the assessment activities. After completion, the motion data are transferred to the Kinesia Web Portal, a secure cloud database, where algorithms are utilized to determine severity scores ranging from 0 (no symptoms) to 4 (severe signs), which have been previously connected with clinical assessments. Severity scores are determined by algorithms when the data are uploaded to a secure cloud repository. Two wireless motion sensors are used by the Kinesia360TM gadget to measure body position, steps taken throughout routine tasks, bradykinesia, dyskinesia, and tremors. The device was made user-friendly by increasing the patient adherence to the video-guided motor activities used to evaluate bradykinesia and tremors in response to drug usage. Numerous investigations demonstrated the device's dependability, with good sensitivity and specificity [21–23,30,33,35,61–72].

The most recent iteration of the DynaPort Hybrid technology, the DynaPort7TM, evaluates movement by integrating gyroscopes and acceleration sensors. It evaluates a person's movement at 100 Hz per channel using a combination of gyroscopes and acceleration sensors. DynaPort communicates with the host PC using the Bluetooth protocol. The phase coordination index, which gauges the accuracy and consistency of left–right bilateral coordination during walking, the total number of walking bouts, the total number of steps, the median walking bout duration, the median number of steps, the variability of the gait pattern, the rhythmicity and smoothness of the gait, the median cadence per bout, the percentage of time spent walking, and the total walking in relation to the total walking and non-walking activity are examples of the quantity measurements. Among other things, it gauges gait rhythmicity, steps, and walking patterns. The DynaPort7 was proven to accurately identify misstep occurrences in a huge cohort of senior adults who did not have dementia or PD [73]. According to in-hospital evaluations, MDS-UPDRS part III was substantially correlated with seven mobility parameters, while the device was acceptable for PD patients [74].

The FeetMe® Monitor is a real-time monitoring device that measures gait parameters and plantar pressure on the insole. This insole has 18 capacitive cell pressure sensors, each with a cell size of around 15 mm2, and one IMU with a 3D accelerometer and a 3D gyroscope to record movement in six directions. Key gait cycle characteristics, specifically heel strike (HS) and toe off (TO) events, are identified by pressure sensors in the insoles, and the associated temporal parameters are computed from the timings of HS and TO events. Each insole calculates the HS, TO, and stride length, which are parameters that depend on the insole. These parameters are sent via Bluetooth Low Energy to a mobile application for every insole. On the FeetMe mobility dashboard, the mobile application enables the real-time recording and/or display of the gait metrics, including actimetry fluctuations and reports on specific gait parameters. The report is tailored to each study and includes both the immediate values on the graphs and the average values over a number of days of examination. The standard gait analysis parameters are as follows: walking dynamics, variability, pace, posture control, and rhythm asymmetry. The performance may be compared with data from healthy people in the same age range after the test results (such as 6 min) are presented. The device was proven effective in PD patients [37,75–78].

The Mobility Lab™ device uses wearable inertial sensors to monitor the gait and balance in patients with PD. It includes six wireless sensors, software for user and patient guidance, and automated analysis. Without doing any particular tests, the user can also hold them for a number of hours. But in this instance, there is no comparison with a normative performance—just raw data in an Excel format. On the other hand, such sensors are frequently employed in a research context to carry out the aforementioned test, maybe in a longitudinal manner. Depending on the appropriate test and measure, patients may employ one to six Opal TMs simultaneously: (i) two sensors on the feet for gait measurements; (ii) two sensors on the arms and one on the sternum for arm range of motion and turning during walking; (iii) one posterior sensor on the lower back (at the L5 level) for balance and postural transition measures. The system can detect a subtle parkinsonian gait in early PD, evaluate parkinsonian gait/posture response to levodopa, and evaluate global gait and balance impairment. It also records FoG episodes with high agreement with clinical raters, making it a valuable tool for the gait and postural analysis of hospitalized patients in an Excel table, which needs qualified medical professionals to interpret [79].

3.2. Acoustic, Optical, Electrical, and Electrochemical Sensors

Acoustic sensors can analyze speech patterns to assess vocal function and speech intelligibility in PD. Xiong et al. [34] analyzed the vocal features of PD-affected individuals with sophisticated computational ML models. The adaptive WSO-sparse autoencoder-LDA model attained better results on a vocal dataset and outperformed the benchmarked algorithms (Logistic Regression, SVM, Random Forest (RF), Naïve Bayes (NB), Gradient Boosting Model). Wan et al. [31] used smartphone sensor data from early PD patients' persistent vowel phonations to predict the severity of the disease with the deep Multi-Layer Perceptron classification algorithm to perform better than other algorithms (including k-Nearest Neighbor (KNN), RF, Linear Regression, M5P regression tree). Deng et al. [80] showed that PD may be more effectively identified using voice, gait/rest, accelerometer, and coordinate tapping data through deep learning models than through self-reports. Interestingly, sustained vowels increased these data's discriminative power [81]. These results indicate an integrative digital biomarker approach for the effectiveness of speech treatment, monitoring the course of the condition, and improving communication techniques in PD by tracking voice characteristics over time. Optical sensors may detect even subtle gait parameters and finger tapping in PD, enabling healthcare professionals to monitor treatment strategies and adjust accordingly. According to Buongiorno et al. [82], an automatic vision system utilizing the Kinect v2 sensor in conjunction with the chosen extracted features may be a useful instrument to assist in the evaluation of postural and spatiotemporal traits obtained from gait, finger, and foot tapping in PD participants. In fact, the SVM-based ML model with foot tapping features reached the best score in terms of 87.1% accuracy, 86.0% specificity, and 87.7% sensitivity. EMG sensors can assess muscle activity and movement patterns in PD, identifying muscle groups affected by the disease and guiding personalized rehabilitation strategies. Advances in electroencephalographic (EEG) technology have enabled the use of these sensors to monitor non-motor symptoms in PD, such as impulse control disorders and heart electrical activity. These sensors may help healthcare providers assess cardiac function and intervene early to mitigate adverse events.

An electrochemical biosensor is defined as "a self-contained integrated device able to provide specific quantitative or semi-quantitative analytical information, by using a biological recognition element (i.e., biochemical receptor), retained in direct spatial contact with an electrochemical transduction element" by the International Union of Pure and Applied Chemistry (IUPAC) [32]. Direct adhesion of an aptamer onto the electrochemically reduced graphene oxide (ERGO) and effective electrocatalytic ERGO enrichment on the electrode may produce enhanced electrochemical signals for the ultrasensitive detection of the serum α-Syn oligomer [83]. The sensor offers a promising tool for early PD diagnosis, with its high sensitivity and outstanding performance for α-Syn oligomer identification.

Rapid quantification and real-time monitoring are still challenges, despite the fact that the severity of motor symptoms is correlated with the plasma levodopa concentration, which was recorded by an Android custom application [84]. These encouraging investigation results unquestionably call for more trials and advancement, along with possible uses in different biofluids, including urine, tears, and saliva.

3.3. Smartphones

Smartphones' extensive use, versatility, and multitasking make them potentially essential tools for identifying common symptoms of PD [85]. To investigate fine motor deficits in early PD, for instance, Iakovakis et al. [86] employed natural smartphone touchscreen typing. A remote study with 253 individuals who contributed 252,000 typing sessions through the iPrognosis mobile app (Apple Computer Inc., Cupertino, CA, USA) and a dataset of 36,000 typing sessions from 39 subjects were used to validate this methodology.

The findings were noteworthy and showed a close relationship with MDS-UPDRS part III. Likewise, digital assessments and traditional clinical scales, such as the MDS-UPDRS, Symbol Digit Modalities Test, and Montreal Cognitive Assessment, demonstrated a good connection in the 12-month WATCH-PD multicenter longitudinal trial conducted with cellphones [54]. Zhan et al. [87] developed and proved construct validity of the mobile PD severity score based on smartphone assessments using a unique machine learning technique. By offering regular, impartial, real-world evaluations that potentially improve clinical care and the assessment of innovative treatments, this score supplements conventional PD measurements. Nevertheless, we are unable to encourage the widespread use of these applications due to the low methodological quality of the identified research, even with the evidence that is currently available.

3.4. Smart Domotics

The word "domotics" is derived from the Latin "domus", implying "house", and "tics", which refers to computer science, robotics, and telematics. In addition to providing overall convenience, safety, and autonomy, domotics and smart homes can be a rich source of real-time data for ongoing monitoring and management of PD [88]. Combining wearable technology with stationary equipment, such as video and temperature, pressure, and movement sensors, may have benefits in thoroughly describing patterns of motion in the home [89]. Furthermore, smart mattresses, automated electrical equipment, and voice-activated lights provide real advantages to those with incapacitating illnesses [90]. Devices for virtual reality and leisure time could be employed in home physiotherapy to help people with PD with their balance and walking. For instance, a recent study conducted at home with a Nintendo® Wii console and wireless controller found that constant visual input made it easier to focus and execute movements [91]. A home-based aerobic exercise gamified program with virtual reality software reduced off-state motor signs in patients with PD of mild severity [92]. Lopez et al. [93] investigated whether smart TV applications were accepted and found that they effectively enhanced IwPDs' cognitive capacities. Yuan et al. [94] noticed that training with virtual reality (VR) was successful at enhancing confidence in preventing falls for IwPDs because it improved posture and balance. Tele-rehabilitation with a treadmill incorporating virtual reality for patients with PD in their homes enhanced their movement and adherence to the training [95].

3.5. Flexible Sensors/Electronic Skin

Ancans et al. [96] created a prototype interactive jacket with nine sensor/actuator nodes linked by three wires for power and data transfer to a wireless gateway using Universal Asynchronous Receiver–Transmitter (UART) hardware in order to close this gap. Human–machine interactions (HMIs) for robot control are made possible by the integration of a bionic crack-spring fiber sensor (CSFS) into the fabric, which may enable the motion monitoring of PD patients [97]. The proposed system comprises a wearable CSFS, a communication module, an NB-IoT base station, a cloud data center, and a mobile terminal. Along with the incorporation of the CSFS, a 5G Narrowband Internet of Things (NB-IoT) system was designed for the collection, delivery, and reproduction of human healthcare data, which demonstrated the system's enormous potential in intelligent telemedicine rehabilitation and diagnosis monitoring, as well as human–machine communication interfaces. In contrast with conventional wearables, more sophisticated neurological E-skin can perceive force, temperature, and light, among other inputs, while retaining the flexibility and softness of biological synapses. The development ofE-skins for intelligent robot applications may be aided by a soft multifunctional neurological E-skin(SMNE) [97]. This SMNE comprises a poly(3-hexylthiophene) (P3HT) nanofiber polymer semiconductor-based stretchable synap-

tic transistor and multiple soft artificial sensory receptors, which are capable of effectively perceiving force, thermal, and light stimuli, thereby better meeting the needs of PD patients.

Triboelectric nanogenerators(TENGs) offer an extra alternative for ongoing PD monitoring based on EcoflexTM and PEDOT by A. Vera et al. [98]. To identify hand motions(such as tremors, tapping of the fingers, or stiffness),the sensor is positioned on the forearm. Additionally, Kim et al. [99] used ocean diatom, chitosan, and catechol to create a stretchy TENG to track PD. On a M-shaped Kapton support, the Catechol–Chitosan–Diatom Hydrogel (CCDHG)-TENG is applied. The resultant sensor is placed on the patient's wrist, and an ML algorithm processes the voltage signal to ascertain the patient's PD status. In order to assess the severity of PD, this study proposed a small, non-invasive smart glove that uses sophisticated, commercial sensors. The gadget is made up of a flexible support that can be placed on the hand and is fitted with inertial and piezoelectric sensors. These sensors are used to collect and process data on hand tremors onboard, giving immediate indicators of the patient's condition. The gadget is based on thin and flexible piezo electric sensors made of aluminum nickel (AlN), which has a high sensitivity for sensing body motions and whose production method is patented by the Italian Institute of Technology (IIT) [100]. The flexible support incorporates two piezo electric sensors: one close to the thumb–index finger joint and another close to the index finger. Three embedded ML algorithms based on NN were developed to classify the acquired piezoelectric and inertial data, each corresponding to standard tests, i.e., finger-tapping, hand gesture, and resting tremor. The test results indicated that the developed embedded models obtained an optimal performance, as witnessed by the high obtained accuracy:95.12%, 98.39%, and 96.62% for finger-tapping, hand-fist closure, and resting tremor, respectively.

4. Discussion

4.1. Benefits

There are several key benefits to using wearable sensors in the management of PD. First, wearable sensors enable the continuous and objective monitoring of motor/non-motorsymptoms, offering a more accurate assessment of disease progression compared with traditional episodic evaluations [12–19,40,43,46,57,59,71,72,74]. Ultimately, wearables are suitable for measuring the autonomic status in real time, such as with ingestible respiratory function devices and cuffless blood pressure monitors. The inter-rater variability, symptom swings in and out of clinic, sensitivity, infrequent rating scales, and questionnaires are some of the shortcomings these solutions address with relation to discrepancies [14]. This real-time data can help clinicians personalize treatment plans, adjust medication dosages, and optimize therapeutic interventions to improve patient outcomes in and out of the clinical setting [53,55,93]. One example of wearable technology is a non-invasive vibrotactile glove that uses implanted mechanical triggers to stimulate PD patients' fingertips and reduce motor symptoms, including tremors and gait [93]. Additionally, wearable sensors can detect subtle changes in motor symptoms that may not be apparent during clinic visits, providing early indicators of disease progression or treatment response [22,30,33,62,63,65]. Physiology-driven inputs from wearable technology can be utilized to guide customized PD treatments, like deep brain stimulation (DBS). More precisely, PD patients' shank velocity from IMUs can be utilized as a kinematic control variable for directing dynamic STN-DBS intervention in real-time to ameliorate freezing and gait dysfunction [92]. Furthermore, wearable sensors enable remote monitoring and telemedicine, allowing patients to receive care from the comfort of their homes and reducing the burden of frequent hospital visits [14].

4.2. Challenges

Despite the numerous benefits of wearable sensors, there are also several challenges and limitations to consider. One of the main challenges is the integration of sensor data into clinical practice, as clinicians may not have the resources or expertise to interpret and utilize the complex data generated by wearable sensors. Khalil et al. [101] employed a single sensor placed on participants' lower backs as a first step toward facilitating mobility testing. Next, the sensor data obtained from participants completing a single mobility task in a typical clinic corridor can successfully distinguish between controls and those with PD. Notably, the manual segmentation and intricate computations of kinesiological characteristics that have been common in previous research [68,78,87] might not be required. Rather than sacrificing the prediction accuracy, algorithmic segmentation, which is followed by the computation of a limited number of sensor-derived characteristics from a single sensor and a single mobility job, provides significant simplicity of data processing and mobility testing. An examination of feature significance showed that timed-up-and-go (TUG) activities, such as turning, walking, turning, and sit-to-stand and stand-to-sit transitions, provided our models with more useful data than more straightforward mobility tests, like gait and sway evaluations, based on an important analysis. These findings support the incorporation of wearable sensors into clinical practice by indicating that significant simplification of mobility testing using a wearable sensor and consequent data processing is feasible without compromising the diagnostic accuracy for PD.In this sense, the use of Socially Assistive Robots (SARs) could help improve the assessment of patient performance, e.g., cognitive tests, at the same time relieving physiotherapists from time-consuming tasks [102].

In addition, issues related to data privacy, security, and regulatory compliance must be carefully addressed to ensure the ethical use of sensor data in healthcare settings. In order to promote a more knowledgeable and involved approach to healthcare decision-making, this involves educating the patients and their caregivers on the advantages, dangers, and moral ramifications of AI technology [103]. Ethical aspects should be incorporated into the design and development process from the outset, for example, in interdisciplinary teams [104]. Enhancing data ownership and accessibility, encouraging open-source solutions, and empowering patients to give their approval for potential applications of their data that serve their interests—for example, by restricting corporate exclusivity and adopting models of meta consent—should all be addressed [89].

Furthermore, wearable sensors may not be suitable for all PD patients, particularly those with cognitive impairments or mobility issues that prevent them from wearing or using the devices effectively. Notably, a decreased score on these smartphone cognitive tests was associated with a lower score on the traditional cognitive assessment [14]. Finally, there is a need for the standardization of sensor technologies, data analytics, and outcome measures to facilitate the adoption of wearable sensors in routine clinical practice.

In practice, it would be reasonable to utilize Personal KinetiGraph to track patient symptoms and adjust therapy as needed. The data supporting the clinical utility of STAT-ON, Kinesia360, KinesiaU, and PD Monitor are insufficient. With incremental cost-effectiveness ratios over GBP 30,000 per quality-adjusted life-year under a variety of potential assumptions, the cost-effectiveness of remote monitoring seems to be generally unfavorable. The longevity of patient symptom improvements was the primary factor that influenced the cost-effectiveness. When compared with the existing level of care, none of the remote monitoring devices offered acceptable value for money [98].

4.3. Future Directions

The future of wearable sensors in the management of PD holds great promise for advancing precision medicine and personalized care. With the continued development of

artificial intelligence and machine learning algorithms, wearable sensors have the potential to predict disease progression, optimize treatment regimens, and identify novel biomarkers for early diagnosis and intervention [24–26,28,29,36,38,41,42,93–110]. Data enhancement and domain adjustment might enhance the generalizability of models in the healthy-to-PD transition scenario, and deep neural network models based on healthy data may be able to effectively identify actions carried out by PD patients [89]. However, the recent PD deep learning models are inadequate against optimal dataset management problems, effective time series data analysis problems, real-time sensor handling problems, data regularization issues, and home-based PD monitoring solutions [91,92]. To resolve these issues, pant body sensor networks are the driving force behind the novel deep clustering model's autonomous health monitoring and analysis paradigm, which enables quick and efficient decision-making [96]. Wider acceptance is restricted by the fact that many commercial goods employ wired connections without revealing technical information.

Additionally, the integration of wearable sensors with smart devices and mobile applications can enhance patient engagement, self-management, and adherence to treatment plans [24–26,28,29,36,38,41,42]. A current study using IMUs combined with convolutional neural network modeling to measure FoG in PD individuals evaluated sensor sites that optimized patient adherence for individualized at-home motor symptom tracking [110]. Collaborative efforts between researchers, clinicians, industry partners, and regulatory agencies will be essential to drive the innovation and adoption of wearable sensors in PD care.

5. Conclusions

Wearable sensors represent a powerful and innovative tool for monitoring and managing motor symptoms in PD. By providing objective, continuous, and remote assessment of key motor symptoms, wearable sensors offer numerous benefits for clinicians, patients, and caregivers. However, the widespread adoption of wearable sensors in clinical practice will require addressing challenges related to data interpretation, privacy, compliance, and standardization. Moving forward, collaborative efforts and interdisciplinary approaches will be essential to leverage the full potential of wearable sensors in improving the outcomes and quality of life for PD patients. Future prospects for wearable sensors in PD include the development of more advanced and user-friendly devices; integration with other technologies, such as artificial intelligence and telemedicine; and validation in large-scale clinical trials. These advancements could revolutionize the management of PD by enabling early diagnosis, personalized treatment plans, and improved quality of life.

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